BLDC Motor Design

ECE 3332

Group #13

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Introduction

A brushless DC (BLDC) motor, also called an electronically commutated synchronous motor, is a type of motor that's built to be more efficient and reliable. It has a rotor with permanent magnets and a stator with coil windings, which eliminate the need for brushes or a mechanical commutator. This design eliminates sparking and reduces wear and tear, making the motor longer lasting and less prone to maintenance issues. Instead of relying on mechanical parts, BLDC motors use electronic controllers to power specific coils at just the right time based on the rotor's position. This creates a rotating magnetic field that keeps the rotor spinning. Thanks to their high efficiency, durability, and compact size, BLDC motors are commonly used in everything from industrial machines and consumer gadgets to electric vehicles and robots.

This project is about designing and building a BLDC motor while staying within specific limits like voltage, current, size and the material provided. Since it is an open-ended project, there is room to get creative and create a motor design that is both smooth and efficient. The goal of this project is also to gain a deeper understanding of electromechanical design and getting hands-on experience with tools such as CAD modeling and 3D printing. In the end, this project connects theory to real-world engineering, helping to build practical skills and spark innovation.

Design Objectives and Specifications

Specifications:

These are the following design specifications we made for our BLDC Motor:

- Rated Torque: 0.23 Nm
- Speed: 2000 RPM = 209.4 rad/s
- Voltage: 12 V
- Current: 4 A
- Power: 48 W

Constraints:

These are the following constraints given for the design of our BLDC motor:

- Power $\leq$ 75 W
- Voltage $\leq$ 15 V
- Current $\leq$ 5 A

- Volume < 84 x 84 x 44 mm (excluding shaft)

Motor Type:

- Radial Flux BLDC Motor → Out-runner
  - # of Teeth: 6
  - # of Poles: 8

Materials and Equipment:

Materials:

- N42 Magnet x 8
- Copper coil (0.42 mm)
- Bearing x 2
- Shaft x 1
- Base x 1

Equipment:

- 3D Printer
- Arduino
- Electric Speed Controller
- Power Supply
- Potentiometer

Design Procedure

Conceptual Design

Selection of Motor Type:

For our motor design, we decided on a radial flux-out runner design due to its high torque density and smooth operation, which aligns with our design objectives. With the motor design, the rotor is on the outside and the stator on the inside, allowing better torque and improved thermal management compared to other BLDC motor designs.

Selection of Poles and Teeth Configuration

For the poles and teeth configuration, we choose 6 stator teeth (a multiple of three) and 8 poles. This combination ensures the number of poles and teeth does not equal each other, preventing motor locking and achieving efficient magnetic interaction. This design selection also aligns with the design principles for 3-phase BLDC motors, ensuring compatibility with our desired winding scheme.

Consideration of the Winding Scheme:

Using the online winding scheme calculator, we determined that the ABC winding scheme in wye (star) configuration was the most suitable. This scheme minimizes cogging, resulting in 24 cogging steps per turn, and enhances the motor's efficiency and smoothness. The calculator confirmed this as the best option for our 6-teeth, 8-pole design. (See Figure X for the calculator output)

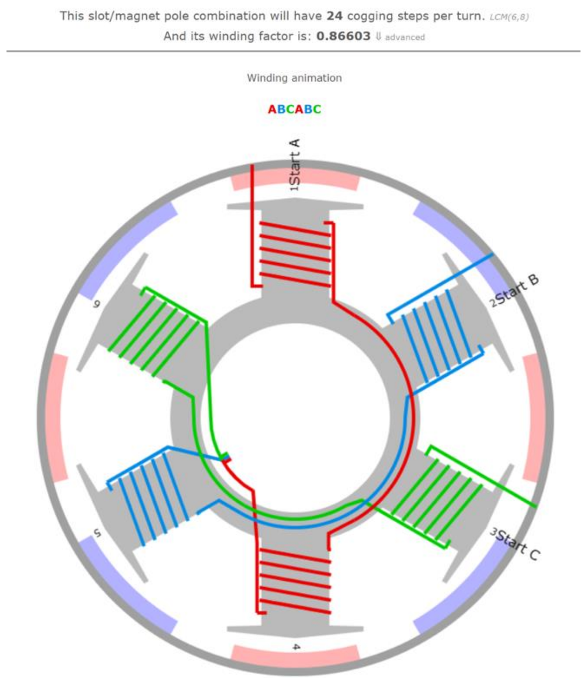

Figure 1: Winding Scheme Calculator

Modeling

The modeling phase began by focusing on the basic shapes and dimensions of the rotor and stator, such as diameter and height to ensure we meet design constraints. The motor was designed to fit within an 84 x 84 x 44 mm box (not including the shaft length) and to align with the three mounting screw holes on the provided base for secure attachment during testing.

Rotor Design:

The rotor design featured magnet cutouts on the inner surface, with a 0.2 mm margin around each magnet to allow for easier fitting while maintaining proper alignment. Additionally, one surface of the rotor was kept completely flat. To simplify 3D printing, one side of the rotor was kept completely flat to ensure proper adhesion to the print bed.

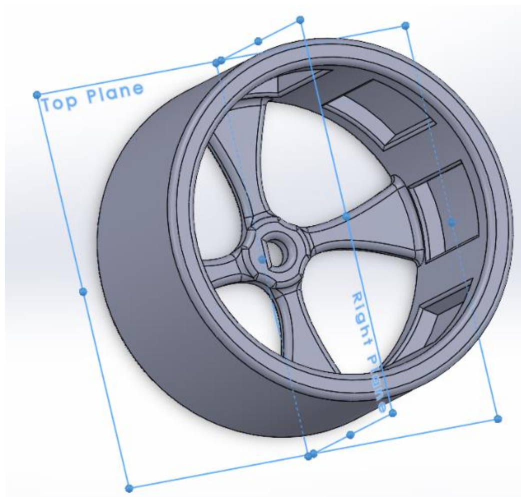

Figure 2: Isometric View of Rotor

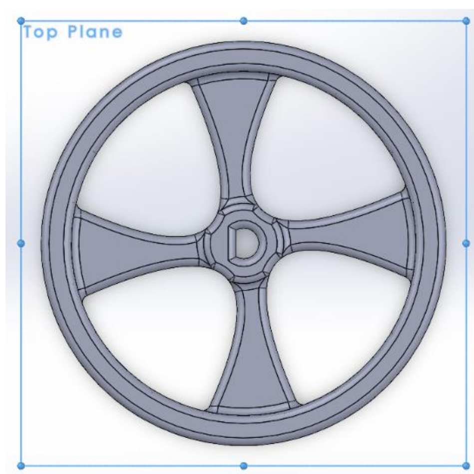

Figure 3: Top View of Rotor

Stator Design:

The stator was divided into two sections to make 3D printing easier, with one surface of each section made perfectly flat. The stator teeth were shaped to minimize the air gap, which was set to 0.5 mm, and to position the coils as close to the magnets as possible, optimizing magnetic flux and overall motor performance. The thickness of the teeth walls was calculated to be 1 mm to ensure efficient flux density. A housing for the bearings was also included, allowing the motor shaft to pass through securely.

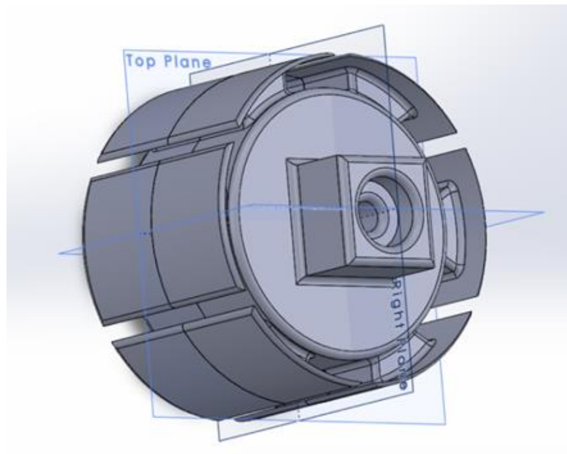

Figure 4: Isometric View of Stator

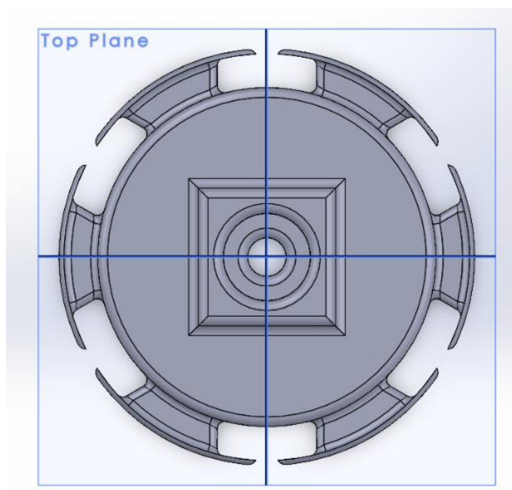

Figure 5: Top view of Stator

Software and File Preparation:

The CAD designs were created using SolidWorks, with each component modeled separately and assembled virtually to ensure compatibility. SolidWorks' assembly tools helped identify and fix any potential design or interference issues, ensuring all parts would fit together smoothly. Once finalized, the CAD files were saved in STL format, ready for 3D printing.

Calculations

Electrical Power P_e:

Where P_e is the input electrical power (W), V is the voltage (V) across the motor and i is the current (A) flowing into the motor

$$P_e = V * i$$

$$P_e = 12\text{V} * 4\text{A}$$

$$P_e = 48\text{W}$$

Mechanical Power P_m:

In an ideal motor, the efficiency is 100%, meaning that all the electric power is converted to mechanical power

$$P_e = P_m$$

Torque Speed Relationship and Mechanical Power P_m:

Where $P_m$is the motor output mechanical power (W), T is output torque (Nm) and ω is the rotational velocity (rad/s) of the motor.

$$P_m = \text{T} * \omega$$

$$T = \frac{P_m}{\omega}$$

$$T = \frac{P_m}{\omega}$$

$$T = \frac{48W}{209.4\, rad/s}$$

$$T = 0.229183118\, Nm$$

Motor Constant K_t:

$$K_t = \frac{V}{\omega} = \frac{T}{i}$$

Flux density at the surface of stator B_a:

Where B_a is flux density (T) at the surface of stator, *t* is the magnet thickness (m), *g* is the radial thickness (m) of air gap, *k* is the thickness (m) of the tooth wall when the material is plastic and B_r is the flux density (T) of magnet

$$B_a = B_r \frac{t}{t + g + k}$$

Based on the research done, B_a should be between 0.8T to 1.2T for the motor design's size. Our target B_a was decided to be 0.8T. Most efficient value air gap for the design was decided to be 0.5mm. The thickness of the N52 magnet is 2mm. B_r is 1.42T according to the chosen specifications for a neodymium magnet of grade N52. Calculations were done to find the most efficient tooth wall thickness for these chosen specifications.

$B_a = 0.8\text{T},\ B_r = 1.42\text{T}, t = 2\text{mm}, g = 0.5\text{mm}$

$$0.8\text{T} = 1.42\text{T}\frac{2\text{mm}}{2\text{mm} + 0.5\text{mm} + k}$$

$$k = 1\text{mm}$$

Number of turns per tooth:

Where *N* is the number of turns per tooth, *T* is torque (Nm), *n* is the number of teeth per phase, B_a flux density (T) at the surface of stator, *L* is the length (m) of stator, *R* is the radius (m) of stator and *i* is the current (A) flowing into the motor

$$N = \frac{T}{4 * n * B_a * L * R * i}$$

$B_a = 0.8\text{T}, T = 0.229\ \text{Nm}, n = 2, L = 29.5\ \text{mm}, R = 21.5\text{mm}, i = 4\text{A}$

$$N = \frac{0.229 \text{ Nm}}{4 * 2 * 0.8\text{T} * 29.5 \text{ mm} * 21.5\text{mm} * 4\text{A}}$$

$$N = 13.96$$

Therefore, the number of windings/turns per tooth is 14.

Assembly

We were given a base, shaft, bearings, copper wires, magnets and we 3D printed the rotor and the stator. The stator's teeth were designed so when we did the windings they would be together instead of far apart. The hole in the stator was designed to be circular instead of the shape of the shaft as we wanted the shaft to spin with the rotor and not cause any rotation to the stator. Small circular incisions were made on both the top and bottom of the stator to hold the bearings. We made the top incision a little shallow so the bearing would slightly stick out which the rotor would sit on not causing friction with the stator while it spun. When assembling, we glued the magnets to the insides of the rotor where we had made incisions of the shape of the magnets. Using the copper wires, we did 14 windings on each tooth and used a wye (star) formation for the wiring connection.

Testing

We designed the motor to be able to handle running at rated values of 48W, 12V and 4A. We used a power supply from a computer and a standard 3 phase brushless speed controller to test our motor before going into the lab. For the first test, we set up the motor with 5V. The motor couldn't start moving on its own, so we gave it a spin by hand which gave it enough momentum to start spinning on its own and we used a speed controller to change our speed. For our second test, we set up the motor to a power supply in the wood shop where we could control the voltage and current, we gave to the motor. We set the values to 2.5A and 12V. Having these values the motor was able to start moving on its own and speed up as well no hand spin required. Through these tests, we could see that our designed motor worked.

Results

We designed the motor to attain a speed of 2000 RPM for our chosen rated values, when testing we did not have the proper equipment to measure the speed, but at the rated values the motor did reach a high speed without breaking.

Conclusion

The goal of this project was to design and build a BLDC motor that ran smoothly and efficiently while staying within the set design limits. We were able to achieve our chosen outcomes in terms of speed and operating power.